

CYTOTOXIC STUDY OF SOME NEW SYNTHETIC DERIVATIVES OF PIPERIDINE

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ABSTRACT

A series of 1-Methyl-4-Hydroxy piperidine with substituted phenacyl halides has been synthesized. These Compounds are evaluated for their cytotoxic activities by Brine shrimp Bioassay and their LD₅₀ was calculated. The SAR studies revealed that derivatives II, V, VII and VIII in which halogen and nitro group in the phenacyl part of molecule may possibly show considerable cytotoxic effects.

INTRODUCTION

Morphine is a representative of natural products which led to the synthesis of several thousands new derivatives having significant pharmacological activities. One of its synthetic derivatives is piperidine based pethidine which possesses lesser side effects than Morphine. A number of derivatives of Morphine have been synthesized uptill now having pharmacological activities. Piperidine nucleus is a common factor in almost all the compounds attributed to exhibit activities like anti-inflammatory and analgesics (Blanchi *et al.*, 1960, Casy *et al.*, 1988 and Feldmann *et al.*, 1991), antidepressant (Iorio *et al.*, 1987 and Wikstorm *et al.*, 1984) and antibacterial (Adams C.P. *et al.*, 1995).

In order to assess the potential of these compounds as effective chemical entity, attempts are made to manipulate the overall nucleus in various ways. In one of the models, phenyl nucleus and hydroxyl groups are placed at the 4th position of the piperidine nucleus and nitrogen atom is substituted with various phenacyl halides in the form of quaternary derivatives (Devinsky *et al.*, and Peter *et al.*, 1986). Phenacyl system imparted a new parameter to the existing piperidine type of compounds, which exhibited analgesic and anesthetic activity (Langlois *et al.*, 1994). A part from analgesics, a wide

range of therapeutic agents are also available which are proved to be cytotoxic (Bold *et al.*, 1985, Horsmann *et al.*, 1987 and 1990).

The inhibitory effects of certain compounds on the proliferation of cancer cells in animals were described half century ago. Since the anti-neoplastic activity of some nicotnamide derivatives have been reported by investigation showed that piperidine derivatives are likely to be the best drug for radiosensitizing the human tumor (Brown *et al.*, 1991). Fischer, Thorston and Coworkers (1994) synthesized substituted piperidine analogs as compound for the treatment of cancer. Another piperidine derivatives was investigated pharmacology as a number of novel class of antitumor agents, exhibited direct and macrophage induced cytotoxicity. Preliminary studies suggest that an undefined cytotoxic effect, macrophage activation and possible affects on signal transduction may account for its anti tumor mechanism of action (Houlihan *et al.*, 1995). In present paper, some new N-substituted Phenacyl derivatives, with piperidine have been reported as new cytotoxic agents. The results of these studies are discussed in this paper.

MATERIALS AND METHODS

Drugs and Chemicals

All the reagents used for synthesis are of analytical grade.

Purity of products are checked on the plate coated with silica gel and spots are visualized under u.v. at 254 nm and 366 nm. Iodine vapours are used to develop the spots. Purity and identity of these compounds are confirmed by UV, NMR, IR and MS.

Melting points are determined on gallenkamp melting point apparatus and are corrected.

Synthetic Procedure

Equimolar quantities of 1-methyl-4-hydroxy piperidine and substituted phenacyl halides are dissolved in acetone separately and then mixed together in a round bottom flask and refluxed on water bath until completion of the reaction.

assayed by brine shrimp bioassay using *Artemia salina*. Results in terms of LD50 in mg/ml are shown in Table-2.

RESULTS AND DISCUSSION

All the compounds (I-IX) are evaluated for their cytotoxicity using Brine Shrimp Bioassay. The approach in this assay is that toxicology is simply pharmacology at a higher dose, thus a simple bioassay might lead to new pharmacologic agent (Mayer et al., 1982 and Dey et al., 1991).

From the data recorded, it is evident that the parent compounds and most of its derivatives have no considerable cytotoxic activity. Only four derivatives (II, V, VII and

Table 1

Comp	R ₁	R ₂	R ₃	R ₄	Salt	Colour/Shape	M.P.°C	Mol. Formula	Yield %
I	H	H	OCH ₃	H	HBr	Colorless crystal	165-166	C ₁₅ H ₂₃ BrNO	64
II	H	H	Br	H	HBr	Colorless rods	260-261	C ₁₄ H ₂₀ Br ₂ NO ₂	72
III	H	OH	OH	H	HCl	Ash white crystal	234-235	C ₁₄ H ₂₁ ClNO ₄	78
IV	H	OCH ₃	H	H	HBr	Yellowish solid	-	C ₁₅ H ₂₃ BrNO ₃	60
V	H	H	NO ₂	H	HCl	Colorless crystal	220-221	C ₁₄ H ₂₁ ClNO ₄	60
VI	OH	H	OH	H	HCl	White Powder	155-158	C ₁₄ H ₂₁ ClNO ₄	65
VII	H	H	Cl	H	HBr	Light Yellow mass	-	C ₁₄ H ₂₀ Br ₂ ClNO ₂	65
VIII	H	H	F	H	HBr	Colorless powder	236-237	C ₁₄ H ₂₀ ClFNO ₂	63
IX	H	H	CH ₃	H	HBr	Brown gummy	-	C ₁₅ H ₂₃ BrNO ₂	72

Conversion of reactions into products is monitored by TLC (CHCl₃: MeOH as solvent system in different combinations). The resulting product is filtered and washed with cold acetone to remove the unreacted starting materials.

Structures and physical properties of these compounds are given in table-1.

Assessment of Cytotoxicity

The cytotoxicity of substituted phenacyl derivatives (I-IX) as well as the parent compound (N-methyl piperidine) has been

VIII) were found to exhibit some degree of toxicity against *Artemia Salina* with LD50 > 1000mg/ml. The rest of five compounds (I, III, I, VI and IX) were devoid of cytotoxicity (LD50 < 1000).

Regarding the structure activity only monohalogenated derivatives (II, VII and VIII) and para nitro moiety (V) showed a low level of cytotoxicity (LD50 886.412, 857.108, 83.636 and 823.238 respectively) suggesting that halogen and nitro group in the phenacyl part of molecule may possible show effect on cytotoxicity.

Table 2
Brine shrimp assay results for (I) and Itsphenacyl derivatives (II-IX)

S. No.	Name of Compound	LD ₅₀ mg/ml
1	1-Methyl-1 (4-methoxy) phenacyl-4 – hydroxy piperidinium bromide (I)	> 1000
2	1-Methyl-1 (4-bromo) phenacyl-4 – hydroxy piperidinium bromide (II)	886.142
3	1-Methyl-1 (3,4 dihydroxy) phenacyl-4 – hydroxy piperidinium chloride (III)	>1000
4	1-Methyl-1 (3-methoxy) phenacyl-4 – hydroxy piperidinium bromide (IV)	>1000
5	1-Methyl-1 (4-nitro) phenacyl-4 – hydroxy piperidinium bromide (V)	823.238
6	1-Methyl-1 (2,4-difluoro) phenacyl-4 – hydroxy piperidinium chloride (VI)	>1000
7	1-Methyl-1 (4-chloro) phenacyl-4 - hydroxy piperidinium bromide (VII)	857.108
8	1-Methyl-1 (4-fluoro) phenacyl-4 - hydroxy piperidinium bromide (VIII)	843.636
9	1-Methyl-1 (4-methyl) phenacyl-4 - hydroxy piperidinium bromide (IX)	>1000

CONCLUSION

Present study reveals that halogenated derivatives of piperidine possess cytotoxic activity to a significant degree. This proves that halogenated compounds of piperidine can impart anti tumor activity. However, further investigation is needed to establish safe and precise but pronounced mechanism of action of the derivatives of piperidine.

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